

PhysioRisk: an interpretable survival-modeling framework for separating demographic and physiological mortality risk

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Abstract

Background: Biomarker-based mortality risk models often show strong predictive performance but remain highly correlated with chronological age, limiting interpretation of whether risk reflects demographic mortality structure or physiological deviation. We developed and evaluated a survival-based framework that separates age- and sex-based demographic mortality risk from an age- and sex-residualized physiological risk component.

Methods: Using NHANES 1999–2016 mortality-linked data, we estimated DemoRisk as the age- and sex-based component of a Cox proportional hazards model with spline-transformed age. Clinical biomarkers were residualized with respect to age and sex, aggregated into physiological axes, and entered into a penalized Cox model to estimate PhysioRisk. TotalRisk was defined as the additive recombination of DemoRisk and PhysioRisk. Performance was evaluated in a temporal test cohort using concordance index, hazard ratios per standard deviation, Kaplan–Meier stratification, and correlation with chronological age. Results were benchmarked against PhenoAge no-CRP proxy (PhenoAge NCP) and PhenoAge NCP acceleration.

Results: In the temporal test cohort (n=6,739; deaths=416), DemoRisk showed strong discrimination but high age correlation. PhenoAge NCP achieved similar discrimination and remained strongly age-correlated. TotalRisk matched comparator discrimination while preserving separation of demographic and physiological components. PhysioRisk showed modest standalone discrimination but low age correlation and a stronger hazard gradient than PhenoAge NCP acceleration.

Conclusions: Mortality risk in NHANES can be separated into a dominant demographic component and a mortality-relevant physiological component derived from age- and sex-residualized biomarkers. PhysioRisk is not intended to replace established mortality-risk scores, but to provide an interpretable decomposition of demographic and physiological contributions to risk.

Keywords: mortality prediction; survival analysis; Cox proportional hazards; interpretability; biomarker modeling; NHANES; prognostic models; risk decomposition

Introduction

Prognostic models for all-cause mortality often achieve strong discrimination by incorporating chronological age and routine clinical biomarkers into a single aggregate score. Although useful for risk ranking, such models can be difficult to interpret because demographic mortality structure and residual physiological variation are combined in the same predictor. This is especially important in aging-related applications, where biomarker-based scores are often interpreted not only as mortality-risk predictors but also as indicators of biological or physiological aging.

A major advance was the development of PhenoAge [1], which directly optimized mortality prediction using clinical biomarkers. While effective, such models typically incorporate age as a dominant feature, limiting interpretability. Epigenetic clocks such as GrimAge [2] further improve predictive performance but at the cost of reduced physiological transparency.

More broadly, prior work on physiological dysregulation has emphasized that aging reflects multi-system deviation rather than a single biomarker signal [3, 4]. In parallel, classical frailty models represent unobserved heterogeneity in mortality risk as a latent multiplicative factor [5], but do not directly map this heterogeneity to observable physiological measures.

Together, these approaches highlight a key limitation: mortality prediction models typically conflate two distinct components of risk—age- and sex-based demographic risk and excess risk arising from physiological variation.

In this study, we propose a survival-based framework that explicitly separates these components. DemoRisk captures the age- and sex-based component of mortality risk, while PhysioRisk represents age- and sex-residualized physiological variation derived from residualized biomarkers. This decomposition enables direct interpretation of mortality-relevant physiological variation after adjustment for age and sex.

The contribution of this study is a transparent decomposition strategy for interpretable survival modeling. Rather than optimizing a single aggregate mortality-risk score, we separate the age- and sex-based Cox linear predictor from a physiological Cox linear predictor derived from age- and sex-residualized biomarkers. This allows total mortality-risk prediction and residual physiological-risk interpretation to be evaluated separately within the same modeling framework.

Risk decomposition framework

Mortality risk can be understood as arising from an underlying hazard function, which integrates over time to determine survival probability [6, 7]. In practice, many biomarker-based “biological age” models achieve strong predictive performance but remain highly correlated with chronological age, suggesting that much of their signal reflects reconstruction of age rather than age- and sex-residualized physiological variation.

Motivated by classical frailty models [5, 8], which decompose hazard into baseline risk and latent heterogeneity, we replace the unobserved frailty term with a measurable physiological proxy. Specifically, we decompose mortality risk into two components: an age- and sex-based demographic risk component, and a physiological component derived from biomarkers after removing age- and sex-associated trends.

This formulation enables explicit separation of age- and sex-related risk from physiological deviation. In the following sections, we formalize this decomposition and demonstrate that the combined model recovers overall predictive performance while preserving interpretability of demographic and physiological contributions.

Methods

Study population

Data were drawn from the National Health and Nutrition Examination Survey (NHANES) 1999–2016 mortality-linked cohort. NHANES is a nationally representative, cross-sectional survey program conducted by the National Center for Health Statistics, with standardized interview, examination, and laboratory components. Mortality follow-up was obtained from the public-use linked mortality files, which provide all-cause mortality status and follow-up time for eligible participants [9, 10]. Analyses were conducted without NHANES survey weights because the objective was internal prediction-model benchmarking and decomposition of risk scores rather than estimation of nationally representative prevalence or population-average associations.

To provide cycle-level transparency, the original NHANES public-use participant counts for each survey cycle included in the analytic window were summarized before application of mortality linkage, adult eligibility, biomarker availability, and benchmark-comparison restrictions. Cycle-level counts were taken from NHANES public documentation and correspond to the number of interviewed participants represented in the public-use survey files, except where CDC documentation separately reports selected, interviewed, and examined counts for later cycles.

NHANES cycle	Original public-use participants / interviewed participants	Examined participants, where reported
1999–2000	9,965	—
2001–2002	11,039	—
2003–2004	10,122	—
2005–2006	10,348	—
2007–2008	10,149	—
2009–2010	10,537	—
2011–2012	9,756	9,338
2013–2014	10,175	9,813
2015–2016	9,971	9,544

Table 1 Original NHANES participant counts by survey cycle included in the analytic window. These counts describe the source NHANES cycles before study-specific exclusions. The benchmark-aligned analytic cohort used for model evaluation was subsequently restricted to adults with mortality linkage and sufficient biomarker availability to construct PhysioRisk and the PhenoAge NCP comparator

Survival time was defined as months from examination to death or censoring using PERMTH_INT, and mortality status was defined using MORTSTAT. Analyses were restricted to adult participants with available mortality linkage and complete data for the biomarker panels required for the benchmark comparison. The analytic window was restricted to NHANES 1999–2016 to preserve alignment with the locked benchmark specification, maintain harmonized biomarker availability across the selected cycles, and avoid reopening model construction after the temporal validation framework was locked. This is consistent with the locked analysis plan, which defines the mortality-linked cohort as NHANES 1999–2016 with PERMTH_INT as survival time and MORTSTAT as the event indicator.

A temporal split was used for model development and validation, with earlier survey cycles assigned to the training cohort and later cycles assigned to the test cohort. This design provides an internal temporal validation rather than random split-sample validation and is reported in accordance with TRIPOD guidance for prognostic prediction models [11].

DemoRisk model

DemoRisk was defined as the demographic component of mortality risk. It was estimated using a Cox proportional hazards model with spline-transformed age and sex as predictors:

DemoRisk ~ cr(age, df = 4) + sex

The Cox proportional hazards framework models the hazard of death as a function of covariates while leaving the baseline hazard unspecified [6, 7]. Age was modeled flexibly using cubic splines because mortality risk increases nonlinearly with age, and age specification is known to materially affect Cox model estimates in epidemiologic cohorts [12].

The DemoRisk model was fit on the training subset only and then applied unchanged to both training and test sets. Penalization was used to stabilize estimation (`penalizer = 0.01`, `l1_ratio = 0.0`). No downstream analyses refit or modified the DemoRisk model. This was a prespecified analysis constraint: DemoRisk was fit once in the training cohort and applied unchanged to all downstream analyses.

Biomarker preprocessing and residualization

To isolate physiological variation not explained by chronological age or sex, each biomarker used in PhysioRisk construction was residualized against spline-transformed age and sex. For each biomarker, a linear regression model was fit in the training cohort:

biomarker ~ cr(age, df = 4) + sex

The residuals from these models represent deviations from expected biomarker values conditional on age and sex. Residualization parameters were estimated in the training cohort only and then applied unchanged to the test cohort. The resulting residuals were standardized using training-derived means and standard deviations.

This procedure was used to reduce age and sex dependence in the physiological predictors before survival modeling. The approach is consistent with the broader goal of distinguishing age- and sex-associated demographic risk from residual physiological heterogeneity, a distinction motivated by frailty models and prior work on physiological dysregulation [3-5].

Axis identification and model selection

To characterize structure within the residualized biomarker matrix, principal components were examined after adjustment for spline-transformed age and sex. These analyses were used for exploratory characterization of latent physiological domains, not as direct model inputs. The leading patterns broadly corresponded to hematologic variation, metabolic processes, and renal/hepatic function, consistent with prior work conceptualizing aging as multi-system physiological dysregulation rather than deterioration of a single biomarker system [3, 4, 13].

Candidate physiological axes were informed by exploratory PCA of age- and sex-residualized biomarkers and by subsequent axis-level ablation analyses. These analyses supported a low-dimensional physiological structure dominated by hematologic stress and renal/hepatic turnover, whereas the metabolic axis contributed less consistently. The final benchmark analysis therefore used a parsimonious two-axis specification, defined before the aligned no-CRP benchmark comparison, with the temporal test cohort used for internal benchmark evaluation of the frozen specification rather than for downstream model refitting. This two-axis specification should be interpreted as a transparent,

parsimonious modeling choice rather than as a definitive biological taxonomy of aging-related physiology.

PhysioRisk construction

The two physiological axes were first constructed as averages of signed, standardized biomarker residuals. HematologicStress included residualized white blood cell count, lymphocyte percentage, mean corpuscular volume, and red cell distribution width; lymphocyte percentage was sign-reversed so that higher values indicated more adverse physiology. RenalHepaticTurnover included residualized creatinine, albumin, and alkaline phosphatase; albumin was sign-reversed. PhysioRisk was then estimated as the linear predictor from a penalized Cox proportional hazards model fit in the training cohort using HematologicStress and RenalHepaticTurnover as predictors. The fitted coefficients were 0.433 for HematologicStress and 0.235 for RenalHepaticTurnover. The resulting log partial hazard was standardized using training-set parameters and applied unchanged to the temporal test cohort. This operationalizes measurable physiological heterogeneity in risk using observed clinical biomarkers, conceptually paralleling frailty models that represent individual heterogeneity in survival analysis [5, 8].

TotalRisk construction

TotalRisk was defined as the additive recombination of the frozen age/sex Cox linear predictor and the standardized physiological Cox linear predictor. This composite preserves the Cox log-risk interpretation of both components while maintaining modular separation between demographic and physiological risk.

TotalRisk = DemoRisk + PhysioRisk

Because Cox linear predictors are additive on the log-hazard scale, this corresponds to a multiplicative decomposition of hazard into demographic risk and residual physiological risk [6, 7]. This formulation preserves interpretability of the two components while allowing their recombination for overall mortality-risk ranking.

PhenoAge no-CRP proxy comparator

A Levine-style mortality predictor was constructed using the published PhenoAge biomarker coefficients, with the CRP term removed. The CRP term was omitted because CRP and high-sensitivity CRP measurements were not consistently available in a harmonized form across the benchmark pipeline, and later NHANES laboratory documentation reflects changes in CRP-related laboratory variables across cycles [14, 15]. The original PhenoAge model was developed as a mortality-trained biomarker-based mortality-risk score using clinical biomarkers and chronological age [1]. Because the C-reactive protein (CRP) term was omitted, this constrained Levine-style comparator is referred to here as **PhenoAge no-CRP proxy (PhenoAge NCP)**. Its age- and sex-adjusted residual is referred to as **PhenoAge NCP acceleration**.

This proxy was used to enable a common benchmark cohort across survey cycles and to avoid introducing additional cohort restriction or assay-related bias from CRP. It should be interpreted as a

constrained Levine-style comparator, not as a full replication of the published PhenoAge model. The benchmark-aligned analytic cohort included 13,090 training participants from NHANES cycles through 2010, with 2,520 deaths, and 6,739 temporal test participants from 2011 onward, with 416 deaths.

PhenoAge NCP acceleration construction

To construct an age-adjusted comparator, **PhenoAge NCP acceleration** was defined as the residual from a train-frozen regression of PhenoAge NCP on spline-transformed age and sex:

PhenoAge NCP $\sim$ **cr(age, df = 4) + sex**

The regression model was fit in the training cohort only and applied unchanged to the test cohort. The residual from this model was used as PhenoAge NCP acceleration. This approach follows the general logic of age-adjusted biological-age acceleration measures while applying the same train-frozen residualization framework used for PhysioRisk [1, 16].

Model evaluation

Model performance was evaluated in the temporal test cohort using four complementary criteria:

- discrimination, assessed using Harrell's concordance index
- effect size, assessed as hazard ratio per one standard deviation increase
- risk stratification, assessed using Kaplan–Meier survival curves across score quartiles
- age dependence, assessed using Pearson correlation with chronological age

The C-index was used to quantify mortality-risk ranking, but it was not treated as the sole performance criterion because concordance can provide an incomplete summary of survival-model performance [17, 18]. Hazard ratios per standard deviation were estimated using Cox models fit to standardized versions of each risk score. These hazard ratios should be interpreted as risk gradients across aggregate scores, not as effects of individual biomarkers.

Kaplan–Meier curves were generated by stratifying participants into quartiles of each score within the temporal test cohort, and survival differences were evaluated using log-rank tests. Kaplan–Meier and log-rank analyses are standard descriptive tools for comparing survival across risk strata [7, 19].

Age dependence was assessed using Pearson correlation between each score and chronological age. This analysis was used to distinguish age-entangled total-risk predictors from age-adjusted physiological-risk signals.

Missing data

The benchmark analysis was conducted on a cohort with sufficient biomarker availability to construct both PhysioRisk and the no-CRP Levine-style comparator. For PhysioRisk construction, analyses used cohort-specific complete-case data for the biomarkers included in the model specification. For the no-CRP comparator, missing biomarker values were handled using training-derived median imputation, with the same imputation parameters applied unchanged to the test cohort.

All preprocessing steps, including residualization, standardization, and imputation where applicable, were estimated using the training data only and then applied unchanged to the test data. This was done to avoid information leakage and preserve the temporal validation structure, consistent with TRIPOD reporting principles [11].

Reporting and validation framework

This study is reported as a model-development study with temporal internal validation, consistent with TRIPOD guidance for prognostic prediction models [11]. It should be interpreted as development and temporal internal validation of an interpretable prognostic modeling framework, not as external validation or clinical deployment of a mortality-risk tool. The temporal split evaluates performance across later NHANES cycles but does not constitute external validation. External validation in independent cohorts is required before clinical deployment or claims of generalizability.

Model formulation and theoretical framework

Mortality risk was represented using the Cox proportional hazards framework, in which the hazard for individual i is

$$h_i(t) = h_0(t) \cdot \exp(\eta_i)$$

where $h_0(t)$ is the unspecified baseline hazard and η_i is the fitted linear predictor [6, 7]. We decomposed the linear predictor into demographic and physiological components:

$$\eta_i = \text{DemoRisk}_i + \text{PhysioRisk}_i$$

DemoRisk was estimated from spline-transformed age and sex, while PhysioRisk was estimated from biomarkers residualized against spline-transformed age and sex. For each biomarker X_{ij} , residualization was performed as

$$X_{ij} = g_j(A_i, S_i) + \epsilon_{ij}$$

where A_i denotes age, S_i denotes sex, and ϵ_{ij} represents the age- and sex-adjusted biomarker residual. The residualized biomarker vector was then aggregated into physiological axes and entered into the Cox model used to estimate PhysioRisk. This formulation is motivated by frailty models that distinguish common hazard structure from individual heterogeneity in mortality risk [5, 8], while operationalizing heterogeneity through observed physiological measurements.

TRIPOD reporting statement

This study was conducted and reported in accordance with TRIPOD guidance for prognostic prediction model studies [11]. The analysis corresponds to model development with temporal internal validation: models were developed in earlier NHANES cycles and evaluated without refitting in later cycles. The primary outcome was all-cause mortality, with follow-up time defined using PERMTH_INT and event

status using MORTSTAT. Predictors included age and sex for DemoRisk, residualized clinical biomarkers for PhysioRisk, and the biomarker set required for the Levine-style no-CRP comparator. All preprocessing steps, including residualization, standardization, and comparator imputation where applicable, were estimated in the training cohort and applied unchanged to the temporal test cohort. Model performance was evaluated using concordance index, hazard ratio per standard deviation, correlation with chronological age, and Kaplan–Meier stratification by score quartile. This temporal split provides internal validation only; external validation is required before claims of generalizability or clinical use.

Results

Primary benchmark comparison

In a benchmark-aligned NHANES cohort restricted to the no-CRP biomarker panel, mortality prediction performance was evaluated across demographic, physiological, and composite risk scores (Table 2). DemoRisk, derived from spline-transformed age and sex, achieved strong discrimination on the temporal test set (C-index ≈ 0.83), consistent with the dominant contribution of age to mortality risk.

Model	Test n	Deaths	C-index	HR per 1 SD (95% CI)	Correlation with age
DemoRisk (age + sex)	6739	416	0.827	3.74 (3.35–4.17)	0.97
PhysioRisk	6739	416	0.654	2.77 (2.66–2.89)	0.11
TotalRisk	6739	416	0.840	1.88 (1.82–1.94)	0.81
PhenoAge NCP	6739	416	0.841	3.98 (3.64–4.35)	0.96
PhenoAge NCP acceleration	6739	416	0.638	1.43 (1.36–1.51)	0.03

Table 2 Benchmark comparison of mortality risk scores in the temporal test cohort

Values are reported in the temporal test cohort of the benchmark-aligned NHANES 1999–2016 analysis. DemoRisk is the age- and sex-based Cox linear predictor. PhysioRisk is the standardized physiological Cox linear predictor derived from HematologicStress and RenalHepaticTurnover after age- and sex-residualization of biomarkers. TotalRisk is the additive recombination of DemoRisk and PhysioRisk. PhenoAge NCP denotes the PhenoAge no-CRP proxy, a constrained Levine-style comparator omitting C-reactive protein. PhenoAge NCP acceleration is its train-frozen age- and sex-adjusted residual

A Levine-style no-CRP comparator (PhenoAge NCP) [1] achieved the highest overall discrimination (C-index ≈ 0.84), but remained strongly correlated with chronological age ($r \approx 0.96$), indicating substantial age entanglement. The recomposed TotalRisk model achieved nearly identical discrimination (C-index ≈ 0.84), demonstrating that combining an explicit age- and sex-based demographic risk component with a residual physiological component can recover the performance of a traditional mortality risk score while preserving separation between demographic and physiological contributions.

Component ablation

To evaluate the contribution of each component, we compared DemoRisk, PhysioRisk, and their recombination as TotalRisk within the temporal test cohort (Table 2). DemoRisk captured most overall discrimination but was strongly correlated with chronological age. PhysioRisk showed lower standalone discrimination but retained low age correlation and a stronger hazard gradient than PhenoAge NCP acceleration. Recombining DemoRisk and PhysioRisk increased discrimination relative to DemoRisk alone while preserving interpretability of separate demographic and physiological components.

Physiology-only comparison

To isolate physiological signal after age and sex adjustment, PhysioRisk was compared to an age-adjusted comparator (PhenoAge NCP acceleration), defined as the residual from a train-frozen regression of the no-CRP proxy on spline-transformed age and sex. Both scores exhibited low correlation with chronological age (PhysioRisk: $r \approx 0.11$; PhenoAge NCP acceleration: $r \approx 0.03$).

Despite similar discrimination (C-index ≈ 0.65 vs. 0.64), PhysioRisk demonstrated a substantially stronger hazard gradient per standard deviation (HR ≈ 2.8 vs. 1.4), indicating greater separation of mortality risk after accounting for age. Kaplan–Meier analyses stratified by quartiles of each score showed clearer survival separation for PhysioRisk, particularly between the highest-risk group and the remainder of the cohort (Fig. 1).

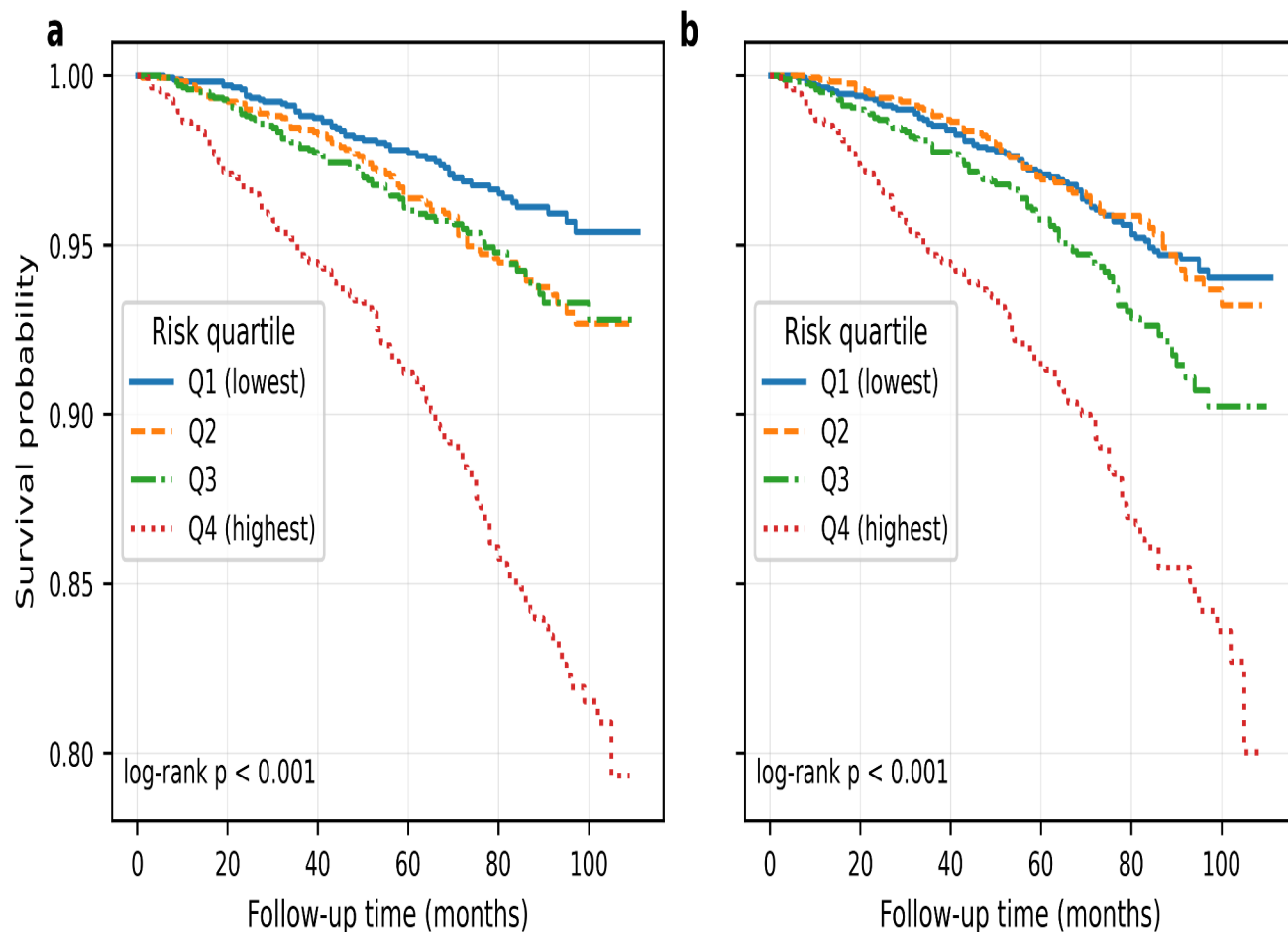


Fig. 1 Kaplan–Meier survival by quartiles of physiology-only risk scores in the temporal test cohort. **a** Survival curves stratified by PhysioRisk quartile. **b** Survival curves stratified by PhenoAge NCP acceleration quartile. PhysioRisk shows clearer separation of the highest-risk quartile, while PhenoAge NCP acceleration shows more gradual separation across quartiles. Quartiles were defined within the temporal test cohort. Differences between quartiles were statistically significant by log-rank test ($p < 0.001$)

Summary of findings

Together, these results indicate that while PhenoAge NCP scores achieve strong overall mortality ranking through age-entangled structure, PhysioRisk showed stronger risk separation than PhenoAge NCP acceleration within this age-adjusted benchmark comparison. Recombining this signal with demographic risk yields a composite model that matches the discrimination of the comparator while preserving interpretability of demographic and physiological contributions. These benchmark patterns are summarized visually in Fig. 2.

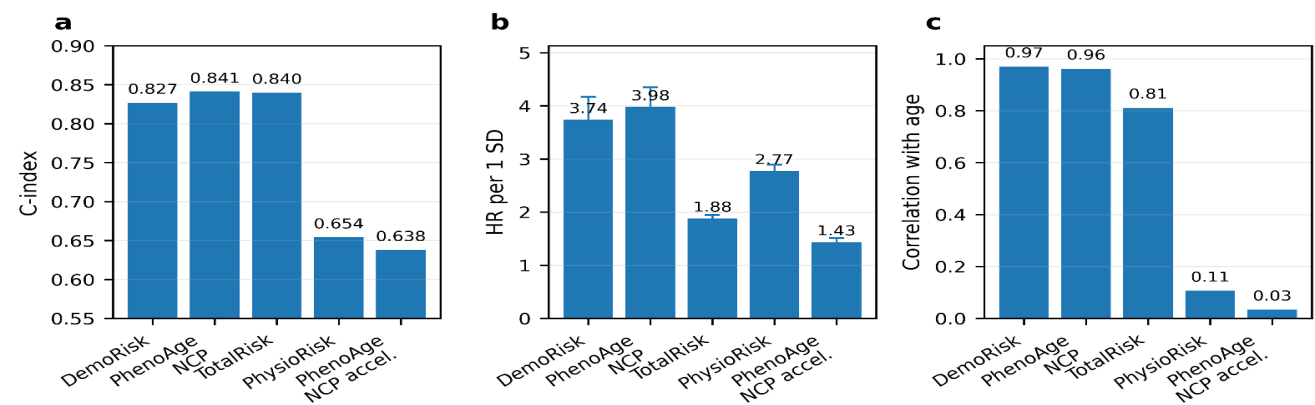


Fig. 2 Comparative performance of mortality risk models in the temporal test cohort.

a Discrimination by C-index. **b** Hazard ratio per one standard deviation increase with 95% confidence intervals. **c** Correlation with chronological age. DemoRisk and PhenoAge NCP achieved high discrimination but were strongly age-correlated. TotalRisk recovered similar discrimination while preserving explicit separation between demographic and physiological components. PhysioRisk showed lower discrimination but reduced age correlation and a larger hazard gradient than PhenoAge NCP acceleration

Discussion

Principal findings

In this study, we developed and evaluated PhysioRisk, a survival-modeling framework that decomposes mortality risk into an age- and sex-dependent demographic component and an age-orthogonal physiological component. This decomposition is motivated by the Cox proportional hazards framework and by classical frailty models, which distinguish baseline hazard from individual heterogeneity in risk [5, 6, 8]. In a temporally split NHANES benchmark cohort, DemoRisk alone achieved strong discrimination, consistent with the dominant role of chronological age in all-cause mortality and with prior work emphasizing the importance of age specification in Cox models [12]. Recombining DemoRisk with PhysioRisk produced a TotalRisk score with discrimination similar to a Levine-style

no-CRP mortality-risk proxy, while preserving explicit separation between the demographic risk component and residual physiological variation.

The physiology-only comparison illustrates the main contribution of the framework. PhysioRisk had lower discrimination than age-entangled total-risk scores, as expected, but retained low correlation with chronological age and showed stronger mortality risk separation than PhenoAge NCP acceleration, the age-adjusted residual form of the no-CRP comparator. These findings suggest that residualized clinical biomarkers contain a measurable mortality-relevant signal that is retained after adjustment for chronological age and sex, consistent with prior work showing that multi-system physiological dysregulation is associated with aging-related outcomes and mortality [3, 4, 13].

Interpretation

The central result is not that PhysioRisk replaces established mortality predictors. Rather, the result is that mortality prediction can be decomposed into two interpretable parts: a dominant age- and sex-based demographic risk component, and a residual physiological component derived from deviations in biomarker values conditional on age and sex. Traditional biomarker-based mortality scores often combine these components into a single predictor. This improves total risk ranking but makes it difficult to determine whether predictive performance reflects age reconstruction, physiological deviation, or both. This distinction is especially relevant because influential biological-age measures have often combined chronological age and biomarkers in ways that produce strong mortality prediction but also substantial age dependence [1, 16, 20].

PhysioRisk makes this distinction explicit. DemoRisk captures the expected mortality gradient associated with age and sex. PhysioRisk captures whether an individual's biomarker profile appears more or less adverse than expected for their demographic baseline. TotalRisk then recombines these components on the log-hazard scale. This structure allows overall discrimination to be recovered while preserving interpretability of the contribution made by physiological deviation.

The comparison with the Levine-style no-CRP proxy should therefore be interpreted narrowly. The no-CRP proxy was used because CRP availability and assay continuity are inconsistent across the NHANES cycles used here. It is not identical to the published PhenoAge model and should not be interpreted as a full replication or as evidence that PhysioRisk is superior to PhenoAge. Instead, it provides a benchmark for testing whether a decomposed demographic-plus-physiology architecture can recover mortality-prediction performance while retaining an age-orthogonal physiological component [1].

Implications

These findings have methodological implications for biological aging and mortality-risk modeling. High discrimination in mortality prediction can be achieved by models that are strongly correlated with chronological age. Such models may be useful for total risk prediction, but their interpretation as measures of physiological deviation is less straightforward. Reporting age correlation alongside discrimination and hazard gradients helps clarify whether a score is functioning primarily as a total-risk predictor or as an age-adjusted physiological signal. This is also consistent with recent methodological

concerns that the concordance index alone is an incomplete summary of survival-model performance [18].

PhysioRisk is best understood as a modular framework for separating these two objectives. TotalRisk is the appropriate score when the goal is overall mortality-risk ranking. PhysioRisk is the appropriate score when the goal is to study age-orthogonal physiological deviation. This distinction may be useful in research settings where the objective is not simply to predict mortality, but to identify physiological heterogeneity among individuals of similar age and sex.

The axis-based construction also provides interpretability. In the present benchmark specification, the strongest and most stable physiological signal was captured by hematologic stress and renal/hepatic turnover axes. These axes should be interpreted as aggregated physiological domains rather than definitive causal mechanisms. Their value in the present study is that they provide a transparent, reproducible way to summarize residual biomarker variation associated with mortality risk. This interpretation aligns with prior work conceptualizing aging as distributed physiological dysregulation across multiple systems rather than deterioration of a single biomarker or organ system [3, 4, 21].

Limitations

Several limitations should temper interpretation. First, this analysis was performed within NHANES using a temporal train/test split. Although this provides out-of-sample evaluation across survey cycles, it remains internal validation within a single data source. External validation in independent cohorts is required before claims of generalizability or clinical utility can be made, consistent with TRIPOD recommendations for transparent reporting and validation of prognostic models [11].

Second, the comparator was a Levine-style no-CRP proxy rather than the full published PhenoAge model. This design choice was made to preserve a consistent benchmark cohort across NHANES cycles and avoid CRP-related availability and assay-harmonization problems. As a result, the comparison evaluates the decomposition framework against a constrained PhenoAge-like benchmark, not against the complete original model [1].

Third, PhysioRisk was developed using predefined physiological axes derived from available clinical biomarkers. Other biomarker panels, axis definitions, nonlinear terms, or interaction structures may yield different performance. The present two-axis model should therefore be viewed as a parsimonious benchmark specification rather than a final biological taxonomy of physiological risk.

Fourth, low correlation with chronological age demonstrates successful statistical age adjustment, but it does not prove causal independence from aging processes. Residualized biomarkers may still reflect age-related biology, cohort effects, unmeasured disease burden, medication use, environmental exposures, or measurement variation. PhysioRisk should therefore be interpreted as age-orthogonal physiological risk, not as a definitive measure of biological age.

Fifth, the current analysis emphasizes relative risk ranking and risk stratification rather than absolute-risk calibration or clinical decision thresholds. Additional work is needed to evaluate absolute calibration, competing risks, decision-curve utility, and performance across demographic and clinical subgroups [22-25].

Finally, regarding survey weights, performance estimates should be interpreted as model-evaluation results within the analytic cohort rather than population-representative estimates.

Future directions

Future work should externally validate the frozen PhysioRisk specification in independent cohorts with mortality follow-up and overlapping clinical biomarkers. Key validation targets include preservation of low age correlation, meaningful hazard gradients, and risk stratification within age and sex strata. Additional work should test whether the framework remains stable under alternative biomarker panels, whether recalibration improves transportability, and whether longitudinal changes in PhysioRisk provide information beyond single-timepoint measurement.

More broadly, the results support a distinction between mortality prediction and physiological-risk decomposition. Rather than treating biological aging scores as single composite indices, future models may benefit from explicitly separating demographic hazard from residual physiological deviation. This decomposition may provide a clearer foundation for studying heterogeneity in aging-related risk and for developing interpretable biomarker-based risk models.

Declarations

Ethics approval and consent to participate

This study used publicly available, de-identified NHANES data and public-use linked mortality files from the National Center for Health Statistics. NHANES protocols were approved by the NCHS Research Ethics Review Board, and participants provided informed consent at the time of data collection. The present analysis was a secondary analysis of de-identified public-use data and did not require additional institutional review board approval.

Consent for publication

Not applicable.

Data availability

NHANES data are publicly available from the National Center for Health Statistics at <https://www.cdc.gov/nchs/nhanes/>. Public-use linked mortality files are available from the NCHS data linkage program at <https://www.cdc.gov/nchs/linked-data/mortality-files/index.html>. The analysis used NHANES 1999–2016 data linked to public-use mortality follow-up.

Code availability

Analysis code used to generate the PhysioRisk scores, benchmark tables, and figures will be deposited in a public GitHub repository before publication. During peer review, code is available from the author upon reasonable request.

Competing interests

The author declares no competing interests.

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Author contributions

conceived the study, developed the modeling framework, conducted the analyses, interpreted the results, and wrote the manuscript.

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